

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0506	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:	T.V.Sai Jagadeesh	Reg. No.:	192472146
Section / Batch:	slot-D/2024	Date:	11/7/26

Code of Conduct

T.V.S. Jagadeesh / 192472146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Sai

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
GRAND TOTAL:			50 Marks

Annexure A – Short Answer Test

1. How does Computer Vision enable computers to interpret and understand visual information?

- a) What are the main objectives of Computer Vision?
- b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

- a) Define feature extraction.
- b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

- a) What is low-level image processing?
- b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

- a) Define a pixel.
- b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

- a) What is the role of an image sensor?
- b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

- a) Define sampling.
- b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

- a) What is perspective projection?
- b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

- a) Define image processing.
- b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

- a) Explain one application in healthcare.
- b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

- a) Why are pixels considered the basic building blocks of digital images?
- b) How do sampling and image resolution influence image analysis?

a) Computer vision is branch of Artificial intelligence, image process, analyze and make decision easily. And also too detect object and recognition.

b) Two - real world application of computer vision are:

→ Face detection

→ Surveillance

2. a) The Feature extraction are:

* Edge

* Smoothing

* Sharpening etc..

b) The object recognition improve pixels values or in computer vision system.

3. a) Low-level is image preprocessing to clear noise removal in the image and also ~~some~~ image enhancement.

b) Difference of mid, high levels:

mid level

Image detect

Some Brighter

High level

Image recognition

Understanding
or
decision

4. a) Pixel:-

It is an picture element and a unit

in the image to make intensive value into digital value.

b) Image Resolution

High Quality

- * More information in image

- * Understandable

- * More pixel value.

And Low Quality

- * Less pixel value

- * Noise is not removal.

- * ~~Not~~ clearly.

5. a) Image sensor:-

It is converting light energy from an object into electrical signal.

Ex:- CCD (charge couple device)
CMOS (complementary metaloxide semiconductor)

b) Image Acquisition Device:-

- * Camera

- * Scanner

6. a) Sampling:-

It is a converting intensity values into the discrete values.

b) Quantization:-

It is a converting intensity values into the digital values.

7. a) Perspective Projection:-

The ~~ima~~ It maps 3D image to 2D image plane, making distant object

b) Illumination provides the light needed for capturing image of greatly image brighter.

8. a) Image Processing:
It is image enhancing or transforming the image using computer algorithm.

b)

Image Processing	Computer vision
→ Enhancement	→ understanding
→ Output of another image	→ Output information decision.

9 a) Application of healthcare is
* X-Ray, * CT scanning etc.
b) Application of Autonomous vehicles
* Traffic sign, etc.

10. a) Pixels store brightness & colour values every digital image is made up of million of pixels.
b) → Sampling determine the number of pixels
→ Resolution determines image detail
→ Higher accuracy and more details.

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Mark
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____